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Digital Twin of an Automotive Brake Pad for Predictive Maintenance

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Abstract

The traditional manufacturing industry is being challenged globally with the comprehensive growth in digital technologies and big data. Digital Twin (DT) technology is one such vision that refers to a comprehensive physical and functional description of a physical component, product or an entire system with all the operational data. The Digital Twin of a product establishes a physical-virtual connection that paves way to real time monitoring all through the entire life cycle of the related product. This paper describes the advance of a digital twin that supports in the predictive maintenance of an automobile brake system. As a proof of concept, brake pressure was measured at different vehicle speeds using ThingWorx Internet of Things (IoT) platform. The data captured using the platform was used to demonstrate the prediction of brake wear using the CAD model implemented in CREO Simulate.

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1. Introduction

1.1. Industry 4.0

The wide spread adoption of communication and information technologies made possible the integration of intelligent components inside the shop floor. This formed the basis for Industry 4.0 which is a strategical response to delocalize production by manufacturing companies [1]. With the key technology areas such as Internet of Things (IOT) and Cyber-Physical Systems (CPS) related to Industry 4.0, manufacturing systems can be detected and measured across wireless networks. This capability has greatly enhanced the gathering and transfer of manufacturing data through the internet. As the data can be exchanged crossways among network infrastructures, it enables a direct amalgamation of the physical components and their virtual systems.

With the practice of these know-hows, Industry 4.0 can open the way for the real time monitoring and harmonization of the manufacturing systems. The real time monitoring of a product will lead to the generation of massive amounts of data which in turn creates a need to utilize Big Data Analytics along the product life cycle. The use of Big Data Analytics is capable of providing personalized and accurate product services as well as improving the quality [2]. The use of Product Life Cycle Management (PLM) was proposed as a data source for Big Data Analytics [3]. The PLM data can also support in the root cause analysis of a product through the integration of maintenance information [4]. It is observed that the current PLM systems are not capable of integrating feedback data throughout the life cycle of a product due to the large amount of data and the difficulty in adapting the available PLM solutions [5]. Adding to the above drawback, PLM systems are also not capable of associating to individual product instances. So, the role of a PLM in the lifecycle of a product is complete when it can receive direct feedback from the field in which the product is under operation. A system concept which integrates IoT and PLM for achieving the same is illustrated in Figure 1.

Nomenclature

w	brake pad wear in mm
k	constant of proportionality ($0.001 \text{ mm}^2/\text{N}$)
P	brake pressure in N/mm^2
v	vehicle velocity in mm/s
t	braking time in s

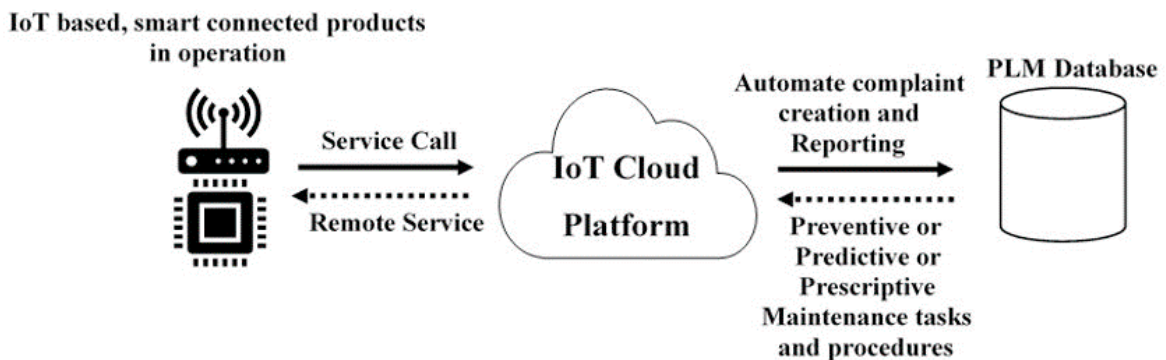


Fig. 1. A system concept for the integration of IoT and PLM

1.2. Digital Twin (DT)

The DT technology was first proposed by Dr. Michael Grieves in 2002 as a PLM concept in a presentation at University of Michigan. The thought of Digital Twin (DT) gained importance as an approach to maintain product related data throughout the entire life cycle. The first definition of DT as stated by NASA is “an integrated multi-physics, multi-scale, probabilistic simulation of a vehicle or system that uses the best available physical models, sensor updates, fleet history, etc., to mirror the life of its flying twin. It is ultra-realistic and may consider one or more important and interdependent vehicle systems” [6]. On a general note, DT can be referred as a virtual depiction of a physical system which is capable of real time monitoring of a product across its life cycle. In literature DT has been proposed for the following categories in manufacturing systems:

- Health monitoring systems: Monitor fatigue, anomalies, deformation and reliability of the system [7-9].
- Digitally reflecting the life of a physical entity to predict its performance affected by different environmental conditions [10,11].
- To optimize a system knowing the historical and current states [12,13].

The above-mentioned applications of DT are primarily targeted at improving performance, monitoring, diagnostics and prognostics activities. It is understood that DT with proper integration of CAD, CAM, CAE, PLM and IoT technologies could pave way for smarter manufacturing systems [14]. The current approach is to develop one such monitoring activity applied to an automotive brake pad to predict brake wear. Further the Digital Twin model can be used to predict maintenance activities based on the measured parameters such as brake pressure and wheel speed.

2. Methodology Overview

2.1. Data acquisition

The process flow of the proposed predictive maintenance system is shown in Figure 2.

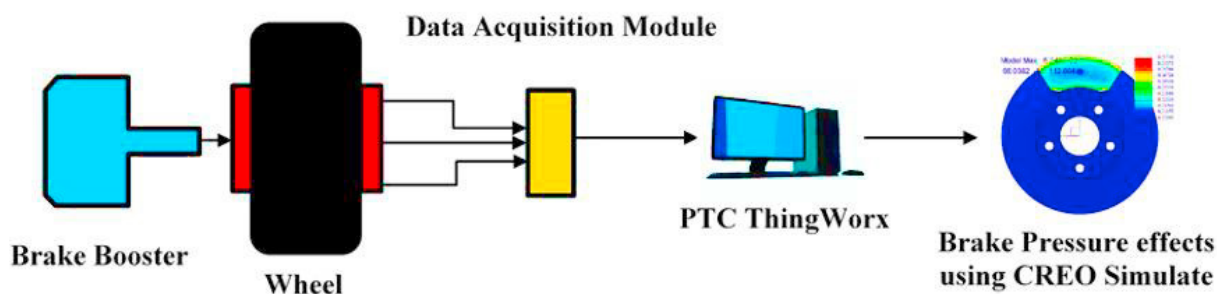


Fig. 2. Schematic of the setup for data acquisition through ThingWorx IoT platform

An automotive braking system setup was prepared using a single wheel and a brake booster of a commercially available passenger vehicle. For the proof of concept, vehicle brake pressure was measured using an appropriate sensor and the data is collected through PTC ThingWorx IoT Platform and the mashup for acquiring the brake pressure data is shown in Figure 3. The different sensor values are imported as “Things” in the ThingWorx platform via Edge Micro Servers. Also, the sampled pressure values acquired at different vehicle speeds are tabulated in Table 1. It is observed that the brake pressure values increase with the increase in vehicle speed.

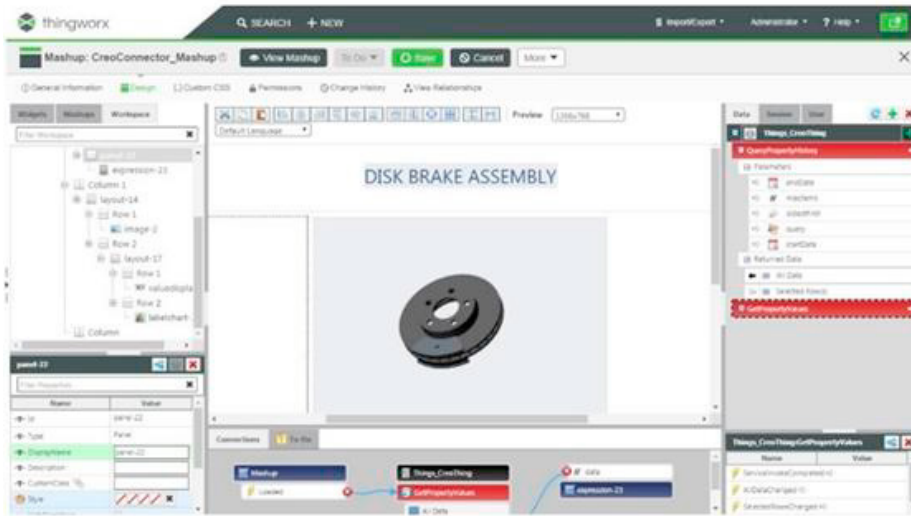


Fig. 3. ThingWorx mashup for the brake pressure data acquisition

Table 1. Sampled brake pressure data acquired for different vehicle speeds

Speed (kph)	Brake Pressure (N/mm ²)
40	1.2
50	2.7
60	4.5
70	6.0
80	7.5
90	8.8
100	10.2
110	11.5

2.2. Brake wear calculation

The acquired brake pressure data is fed into an instrumented assembly modelled in CREO through the 'Connect to server' option. The isometric view of the assembled brake pad and rotor is shown in Figure 4. The modelled brake rotor has a radius of 127.5mm and the brake pad dimensions were matched with the brake pad available in the vehicle.

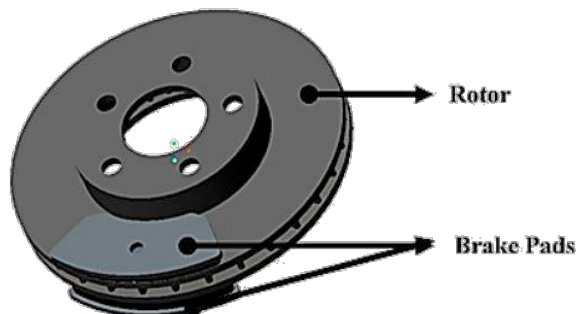


Fig. 4. Isometric view of the instrumented assembly modelled in CREO

With the acquired data, the brake wear is first calculated using an analytical equation as given in [15] and the same is shown as equation (1).

$$w = kPvt \quad (1)$$

After the proper unit conversion and application of equation (1), the brake pad wear in 'mm' for the acquired values were found and the values are shown in Table 2. Theoretically, it is observed that the brake pad wear increases with increase in the vehicle speed and brake pressure.

Table 2. Calculated pad wear for different brake pressure values

Speed(kph)	Brake Pressure (N/mm ²)	Wear (mm)
40	1.2	0.064
50	2.7	0.182
60	4.5	0.364
70	6.0	0.567
80	7.5	0.810
90	8.8	1.069
100	10.2	1.377
110	11.5	1.707

2.3. Brake wear simulation in CREO Simulate

With the acquired data and the modelled assembly, a structural simulation is done in CREO Simulate. The structural boundary conditions for the brake pad and rotor are modelled as shown in Figure 5. The rotor is fixed in all degrees of freedom except in Y-axis allowing to simulate the rotation of a car's wheel. Angular velocity of the rotor in the Y-axis was set to 87.05 rad/s which is equivalent to 40 kph vehicle speed. Similarly, angular velocity values were calculated for the rest of the vehicle speeds shown in Table 1 and were used for the simulation of brake pad wear. A surface-surface bonded contact was used to simulate the friction between the brake pads and the rotor.

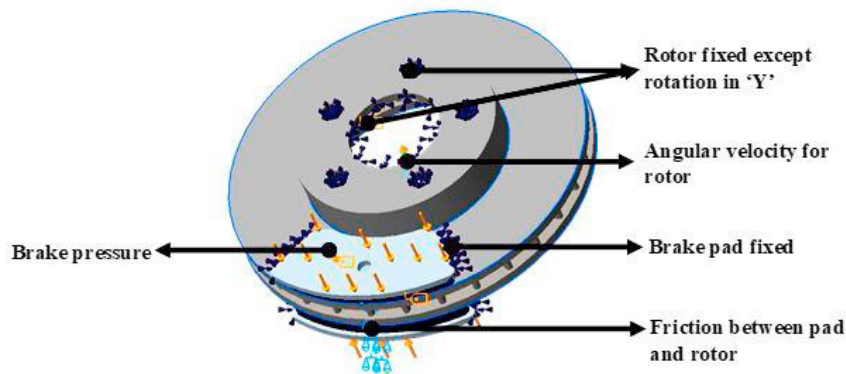


Figure 5. Structural boundary conditions in CREO Simulate

A sample simulation results for the first value of 1.2 N/mm² is displayed in Figure 6. It is observed that the maximum displacement appears at the outer edges of the brake pads. Similarly, the simulation was carried for the rest of the brake pressure values and the obtained results are tabulated in Table.3. The results show a good correlation between the simulated and the calculated values with an average 11% error.

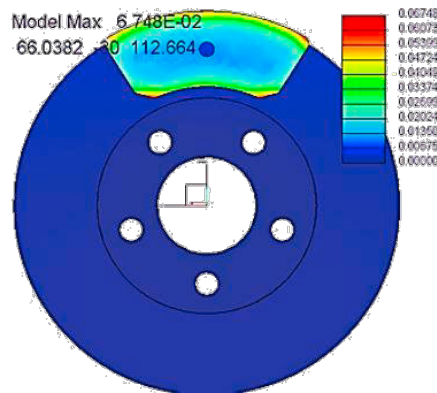
Figure 6. CREO Simulate displacement result for 1.2 N/mm² brake pressure

Table 3. Comparison of the calculated and simulated brake pad wear

Speed(kph)	Brake Pressure (N/mm ²)	Calculated Pad Wear (mm)	Wear from Simulation(mm)
40	1.2	0.06	0.06
50	2.7	0.18	0.15
60	4.5	0.36	0.25
70	6.0	0.57	0.33
80	7.5	0.81	0.72
90	8.8	1.07	0.98
100	10.2	1.38	1.15
110	11.5	1.71	1.61

As the sample size is small, the data was analyzed for single tail ‘t’ test and the ‘p’ value was calculated to value the significance. The t and p values were calculated as -1.126 and 0.1486 respectively using RStudio with a confidence interval of 99%. It is observed that the mean wear value for the sample data is 0.7675mm. The one-sided 99% confidence interval specifies that the mean wear rate is expected to be less than 1.38mm. The p-value of 0.1486 indicates that if the mean wear of the pad is 1mm, the likelihood of picking a sample with a mean wear less than or equal to the p-value would be approximately 18%. Since the p-value is not less than the significance level of 0.01, we cannot reject the null hypothesis that the mean wear value is equal to 1mm. This means that the possibility of wear exceeding 1mm is nil.

3. Results and Discussion

The role of digital twin for the predictive maintenance of an automotive brake pad is described in this paper. The real time brake pad data from a brake pad set up is acquired using ThingWorx IoT platform and the data is connected to CREO taking advantage of the software capability. A digital twin of the brake pad and rotor is modelled in CREO and the brake pad wear is simulated in CREO Simulate. The simulated data is comparable with the calculated pad wear data. The developed system can be used to gather real time data when the actual vehicle is in use. Further, collection of real time data for a longer period will help in big data analytics and provides a chance for prescriptive maintenance of brake pads.

Automotive OEM’s are yet to launch any advanced predictive maintenance solutions for braking systems to the market. Simple maintenance solutions are used where the brake pad usage is monitored and projected into the future based on the data acquired through on-board systems. Volvo has recently implemented a predictive maintenance system wherein diagnostic trouble codes are wirelessly sent to a service location office enabling the inference of a

probable failure cause. With the data set collected with the developed system, machine learning algorithms such as filter and wrapper-based methods can be used to predict brake wear and life [16]. While the current research is limited to the proof of concept of the digital twin capability in the predictive maintenance, the model can be further enhanced to full vehicle digital twin models. Future works will concentrate on the development of a full vehicle digital twin with the support of additional sensors and sub systems.

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